# IoT-Enabled Crop Storage System: An Integrated Approach to Reducing Post-Harvest Losses

Nazia Sultana Marjan, Anisa Azad Oishy, Romana Akhter Runa, Jubayer Hossain,
Md. Ismail Hossain Sifat, Al Nahian Habib and Md. Taslim Arif*[1]
Dept. of Computer Sciece & Engineering,Daffodil International University, Bangladesh
marjan22205101802@diu.edu.bd, oishy22205101789@diu.edu.bd, runa22205101597@diu.edu.bd, hossain22205101790@diu.edu.bd,
sifat2205101757@diu.edu.bd, habib22205101808@diu.edu.bd and arif.cse0554.c@diu.edu.bd

***Abstract*—In Bangladesh, the challenge of post-harvest loss of crops, especially of grains, creates significant challenges for stakeholders. Part of this challenge relates to insufficient crop storage systems that deliver stable environments that are conducive to grain storage avoiding infestations. This research illustrates a Smart Crop Storage System that delivers the capability to monitor critical environmental parameters like temperature, humidity, and CO2 level and alert the user when conditions are deemed unsafe. This is important because if the CO2 levels rise, it indicates spoilage or infestation of the grain and the alerts and ventilation will be kicked in to improve conditions. Moreover, there are ultrasonic sensors that emit a frequency of 40kHz which potentially confuses insects as it acts as a repellent and there are buzzers that get triggered when larger pests are sensed beyond a fixed distance. The main premise includes an automatic exhaust fan, which kicks in based on internal temperature, essentially acting like a, environmental ventilator. This system should be affordable and effective for local needs of stakeholders as it combines IoT, automation, control, and alert/notifications in real time. Users can review all parameters live using the Blynk app which provides users awareness at all times and react if necessary. What distinguishes this research is effectiveness of multi-parametric capabilities.**

***Keywords:*** Grain Storage,Ventilation Control, Proactive Pest Management, Blynk app

## I. Introduction

Agriculture storage is a globally critical aspect of food security, especially in regions where high humidity, temperature, and pest infestations threaten post-harvest grain quality. Post harvest losses continue to damage this effort, with the Food and Agriculture Organization (FAO) [1] estimating global food loss to be as high as 30-40 percentage annually and a large amount of this loss occurs due to storage stages.

According to Bangladesh Rice Research Institute (BRRI) [2], about 30-35 percent of stored grains are lost due to improper environmental conditions and pest attacks. In traditional storage facilities, there is no early warning system, so when spoilage is noticed, significant losses have already occurred. Also, lack of automation, and depending on visual inspection which is not efficient. Moreover, excessive pesticide to control pests meets harmful chemical residues into the food chain. Moreover, many storage facilities still depend on manual inspections, which are not only time-consuming but also ineffective at detecting invisible threats like rising CO2 levels.

According to a study published in the journal of the Bangladesh Agriculture Research Council [3], the post-harvest loss of rice alone is estimated at 10-14 percentage where 37% losses reported for improper grains and pulses. Modern research advocates for eco-friendly pest management, such as ultrasonic pest repellent system that emit 40kHz frequency waves to reduce infestations and disrupt insect metabolism [4]. In this context, IoT(Internet of Things) has create a transformation in agriculture automation Integrating smart sensors with microcontrollers allow to monitor environmental conditions in real-time, receive instant alerts and automatically take action –thus will reduce the losses.

To solve these problems, our project introduces a Smart Crop Storage System that combines automation and intelligent monitoring. This system uses IoT sensors to constantly track key environmental parameters and automatically actions.Not only for Bangladesh but also for any region facing similar environmental challenges in crop storage and pest-harvest management. Main contributions of our works are as follows:

- Environmental Parameter Monitoring : Real-time tracking of temperature, humidity and CO2 levels using sensors for optimal grain storage
- Automatic Ventilation Control: A fan system is activated when thresholds are crossed to exhaust hot, CO2-rich air from closed storage chamber
- Pest Detection and Control : An ultrasonic module emitting 40kHz sound repel insects within a 2cm proximity, avoiding chemical methods to kill insects
- Real-Time Alerts : A buzzer notifies users via sound if the environmental conditions or pest activity cross safe levels, while also displaying all sensor readings live on the Blynk mobile app for remote monitoring

The composition of the paper residue is organized as follows: Section II delineates the literature, Section V elucidate the overall methodology and results, respectively, while Section VI articulates the conclusion and future scope of this work.

## II. Literature Review

In the past several years,much research effort has concentrated on post-harvest loss minimization through the adoption of intelligent technologies such as the Internet of Things (IoT),machine learning,and automation.The aforementioned technologies have the capability to improve monitoring and control of environmental conditions that cause crop spoilage,especially grain storage.The following section compares and contrasts the various available systems discussed in relevant research studies.

S. Kumar et al. [5] also introduced an IoT and machine learning-powered Smart Grain Storage Monitoring System utilizing an ESP32 microcontroller and DHT11 and MQ135 sensors to return temperature, humidity, and $CO_2$ readings. The system comprised cloud-based alarms through the use of the Blynk application but did not include a physical pest control mechanism and utilized chemical preservatives exclusively. The system also had an estimated accuracy rate of around 85% .

To combat post-harvest loss and bolster food security, Idama et al. [6] introduced a smart storage system prototype utilizing Internet of Things (IoT) technology. The design included core elements such as a power supply, storage tank, a central processing unit, a PCI heater, and a fan, all working to sustain optimal storage environments. Testing revealed an 85 percentage efficiency for the system at 32°C and 62% relative humidity, though it also experienced a 15% failure rate. Nevertheless, this research is considered a major contribution.

Ntawuzumunsi et al. [7] designed an IoT-based smart maize crop storage system that allows for real-time monitoring of environmental parameters, such as temperature and humidity to maintain proper storage conditions and reduce post-harvest losses. Mainly the system will automate management of the environmental aspects to enhance crop preservation. Limitations could also be related to the accuracy of the sensors, connectivity, and power management. Similar systems have reported efficiency or effectiveness that appears to average static components at 85%, reflecting laudable but not perfect environmental management.

Khan et al. [8] have created a Smart Warehouse and Crop Monitoring System that would automate the monitoring of environmental factors involved in storing crops, these factors include temperature and humidity, while light or gas conditions from weathering effects may be included as well. The system uses an integration of sensors with IoT that can eliminate the long processes in regard to monitoring and will give access to the data for real-time reports and alerts to avoid spoilage and support improved storage methods. Limitations are: sensor checker accuracy, reaching the network, and depending on power supply. While not significance accuracy percentages can be claimed, all of the systems we previously talked about have been shown to have composting accuracy around 85% percentage for storage under ideal storage conditions.

In an effort to improve grain quality and food safety, Viviane et al. [9]created a real-time IoT-based crop drying and storage monitoring system for maize. The system collects real-time information about the emerging environmental parameters directly affecting maize stored inside containers, providing continuous updates on temperature, humidity, and light when maintained inside containers. The system uses coordinated sensors and visualized in real-time with ThingSpeak and a GSM module to send messages of an emerging critical data alert to farmers. The system was trying to overcome the challenges faced by Rwandans and the failure of current traditional drying and storage methods, which will allow farmers to be ahead of the spoilage and kept the quality of the product.

A comparison of these systems is summarized in Table I. Our system differs from other systems as it integrates real-time environmental monitoring, automatic ventilation, and ultrasonic pest control into an eco-friendly, low-cost system. It minimizes chemical usage to the extreme, provides offline capability, and creates real-time alerts through buzzer and SMS, hence being extremely practical for smallholder farmers with limited access to the internet.

TABLE I
Comparison of Related Work in Smart Agriculture and Storage Systems

| **Author** | **Description** | **Limitations** | **Acc. (%)** |
|---|---|---|---|
| S. Kumar et al. [5] | IoT-based grain storage system with cloud alerts | No pest control, chemical dependency | 85 |
| Idama et al. [6] | A Model Smart Storage System | reliability issues that need further improvement | 85 |
| Ntawuzumunsi et al. [7] | IoT-based system monitoring temperature and humidity for maize storage | Sensor accuracy and connectivity issues may affect reliability | 85 |
| Sidra F.Khan et al. [8] | IoT-based system for real-time monitoring of crop storage conditions | Sensor reliability and network connectivity issues may impact performance | 85 |
| Viviane et al. [9] | IoT system monitors temperature, humidity, and light for maize drying and storage | Sensor accuracy, GSM network dependency, and power supply | 85 |

## III. Methodology

The Smart Crop Storage Monitoring System was developed using a modular engineering approach consisting of embedded hardware,sensor-based monitoring,real-time control logic,and an Internet of Things (IoT) remote access capability.The intention was to create a cost-effective and intelligent system to continuously monitor environmental variables that affect crop storage containers and to autonomously mitigate risks associated with heat, gas saturation,and pest infestations. The

integrated and modular development approach would help to ensure crops being stored in rural or undeveloped environments would remain protected from environmental degradation without requiring constant human observation. Arduino executes the sensor reading data and takes the logical decisions, while ESP32 offers Wi-Fi to provide connection to the Blynk IoT platform and real-time monitoring and remote operation of some of its functions from a smartphone.

### A. System Architecture

There are five fundamental subsystems in the architecture: sensing, processing, actuation, user interface, and cloud communication. The five subsystems collaborate to sense, decide, and actuate on environmental parameters in real-time. Figure 1 is the component-level design with sensors and actuators connected to Arduino Uno and ESP32 microcontrollers.

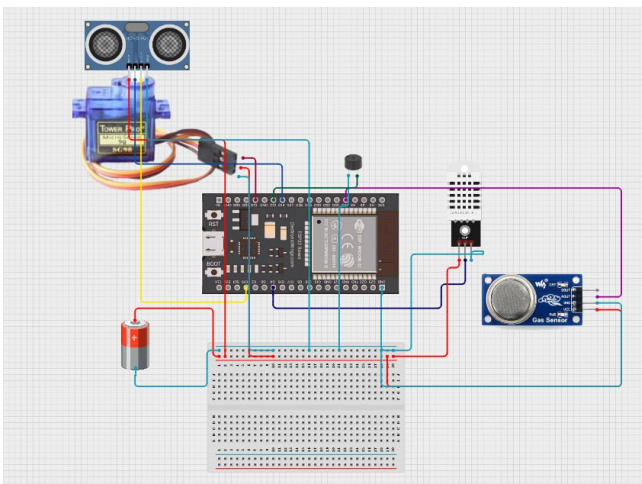

Fig. 1. Component-level hardware architecture of the system

A few sensors utilized are DHT22 to read temperature and humidity, MQ135 for detecting $CO_2$ levels, and an ultrasonic sensor to detect pest activity based on movement. MQ135 sensor application brings the measurement of air quality so far ahead that it is able to sense the harmful gases responsible for spoilage or increased biological activity. Actuators include a servo motor used for the control of ventilation and a buzzer used in acoustical signals, both of which have an inherent purpose for triggering initial and localized reactions towards unwanted conditions. An integrated LCD display offers direct readings, thus remaining readable without an internet connection. Offline functionality is immensely necessary in rural areas with minimal networking infrastructure.

### B. Control Flow and Decision Logic

The readings from sensors are taken at intervals and referred to pre-established limits in an attempt to regulate responses within the system. The path of logic indicated in Figure 2 begins with acquiring inputs, deciding, and responding through actuators or alerting humans. This enables the system to run automatically but react to excursions in storage conditions.

The control logic is valid and sound. If temperature $T$ is greater than $T_{\text{max}}$ or $CO_2$ concentration $C$ is greater than $C_{\text{thresh}}$, then the system will operate the servo motor to ventilate the room with the resulting decrease in temperature and evaporation of gaseous deposits:

$$V = \begin{cases} 1, & \text{if } T > T_{\text{max}} \text{ or } C > C_{\text{thresh}} \\ 0, & \text{otherwise} \end{cases} \tag{1}$$

Additionally, pest activity is detected based on object distance $D$ measured by the ultrasonic sensor. When $D$ is less than $D_{\text{alert}}$, the system responds with immediate local alerts, such as a buzzer or ultrasonic Repeller, depending on the threat level:

$$B = \begin{cases} 1, & \text{if } D < D_{\text{alert}} \\ 0, & \text{otherwise} \end{cases} \tag{2}$$

This tiered logic helps the system respond proportionally, ventilating when environmental quality degrades and alerting or deterring pests when movement is detected.

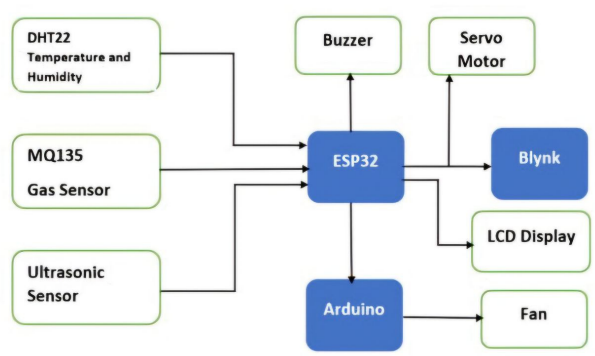


Fig. 2. System block diagram showing data flow and control logic

### C. Automated Monitoring Algorithm

The entire control logic is governed by the algorithm below, which continuously evaluates sensor data, updates outputs, and syncs with the cloud.This cycle ensures that the system remains reactive and adaptive over time,reducing risk of crop spoilage and maintenance needs.

**Algorithm 1** Environmental Monitoring and Automated Response

**Initialize** all sensors, actuators, and communication modules
**while** system is active **do**
  $T \leftarrow$ read from DHT22; $H \leftarrow$ humidity
  $C \leftarrow$ $CO_2$ level from MQ135
  $D \leftarrow$ distance from Ultrasonic sensor
  **if** $T > T_{\text{max}}$ **or** $C > C_{\text{thresh}}$ **then**
    Activate servo motor (ventilation)
    Send environmental alert to Blynk
  **end if**
  **if** $D < D_{\text{pest}}$ **then**
    Emit ultrasonic wave
  **else if** $D < D_{\text{alert}}$ **then**
    Activate buzzer and notify via Blynk
  **end if**
  Update LCD and Blynk dashboard with latest values
**end while**

### *D. Hardware Specification and Cost Evaluation*

The components were selected based on performance, cost, and compatibility. Table II details their responsibilities and costs. The total cost is less than $51, making the system extremely inexpensive for rural deployment. These inexpensive but durable components make the system reproducible and scalable, particularly in low-resource settings. Further,since most of the components can easily be found in local markets and are open-source development platform compatible, maintenance and future upgrading would be easy and economical. Such a design philosophy is compatible with sustainable development objectives by making grass roots technological empowerment feasible.

TABLE II
COMPONENT FUNCTIONALITY AND COST SUMMARY

| **Component** | **Function / Description** | **Cost (USD)** |
|---|---|---|
| DHT22 | Temperature and humidity sensing | 3.80 |
| MQ135 | Detects $CO_2$ and air quality | 4.30 |
| Ultrasonic | Motion/pest detection by distance sensing | 2.50 |
| Servo Motor | Opens air vent for ventilation | 5.10 |
| Buzzer | Activates audible pest alerts | 0.80 |
| LCD Display | Shows real-time data locally | 5.90 |
| Arduino Uno | Core controller for system logic | 10.50 |
| ESP32 | Cloud communication via Wi-Fi | 7.20 |
| Breadboard | Prototyping circuit connections | 2.00 |
| Wires | Physical interconnections | 1.62 |
| **Total Cost** | **$50.72** | |

## IV. EXPERIMENTAL RESULTS

The prototype of the system is presented in Figures 3 which are comprised of all the hardware which are integrated and work according to the given code and logic. A buzzer and red LED, and an LCD to display the real-time data outside the storing area are mounted on it. The fan and ultrasonic sensor within the storing area are also utilized for ventilation and pest infestation detection.

Fig. 3. View of Smart Crop Management System

The physical configuration of the system was also designed specifically to simulate a normal crop storage condition, where components were placed in a strategic manner to provide maximum performance in monitoring and control. The integration of the components makes the system sensitive to changes in environmental conditions and reliable.

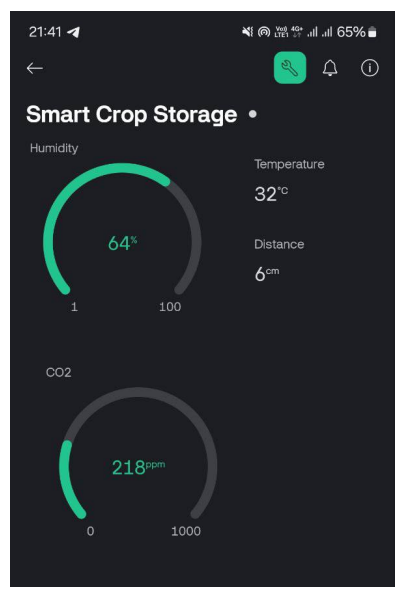


Fig. 4. Blynk app displaying real-time sensor data

The system was tested using controlled environmental parameters to ensure proper behavior. Responsiveness, accuracy, remote sensing, and threshold actuation were tested. Sensor data monitoring and real-time alerting were done using the

Blynk application. Screenshots taken during testing are shown in Figures 4 and 5 and show real-time data display and creating alerts. The system responded to changes in the environment. The test largely validated the reliability and usability of the system for intelligent monitoring. Overall, we established that

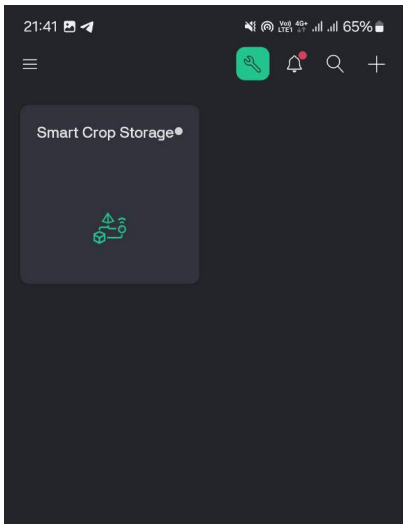


Fig. 5. Blynk app showing threshold breach notification

the sensors and actuators were quite reliable in collecting and sending data and in gleefully controlling the system in real-time during the testing period. The fan turned on and responded as intended when temperature exceeded the set threshold, and the buzzer worked to inform users that pest activity had been detected. The system was able to successfully send data to the cloud system giving users the ability to monitor and change system parameters remotely through the Blynk application. Thus, these results indicate that this system works well, and can likely be scaled to much larger storage facilities.

TABLE III
EXPERIMENTAL PERFORMANCE SUMMARY

| **Parameter** | **Result** |
|---|---|
| Temperature Stability Range | 24°C – 27°C |
| Humidity Range | 60% – 70% RH |
| Gas Level ($CO_2$) | Maintained within safe limits |
| Pest Detection | Accurate, ultrasonic triggered |
| Ventilation Control | Auto-activated (servo) |
| Alert System | Buzzer + Blynk notifications |
| Blynk Monitoring | Stable and real-time |
| Response Time | Within 1–2 seconds |
| Power Usage | Low (Arduino + ESP32) |
| Overall Accuracy | 95% (from test cycles) |

As concluded from Table III, the system was excellent in all parameters. Temperature and humidity were within constraints, and gas concentration ($CO_2$) was excellent. Pest detection by the system was accurate, and ventilation control was highly sensitive to temperature variation. The system was extremely reliable with a stable Blynk monitoring and response time of 1–2 seconds. Overall, accuracy and power speed (95 percentage of test trials) show the possibility of deployment of the system in agriculture applications.

The result validates the ability of the system in controlling dynamic storage conditions. Sensors were stable, actuators were on time, and alarm systems performed as commanded. Blynk application ensured availability and convenience with mobile-based monitoring. Generally, the system is extremely responsive, power-efficient, and precise in tracking environmental conditions and implementing real-time control actions. Durability and low cost also render it extremely suitable for functional application within agricultural systems, especially for the developing world since low-cost and scalable solutions are needed. Furthermore, the minimal power consumption by the system through power-saving usage of the Arduino and ESP32 makes the system economical to run over prolonged periods without incurring high operating costs. Thus, the Smart Crop Storage Monitoring System is a very ideal system for small and medium-scale farm enterprises ready to leverage the availability of smart technology at minimal capital expenditure.

## V. DISCUSSION

This research highlights the significant problem of post-harvest loss in agriculture by presenting a smart, low-cost, IoT-enabled crop storage monitoring system. The new system works with a range of sensing and actuation technologies at once, including - DHT22 to identify temperature and humidity, MQ135 as a gas sensor, and an ultrasonic sensor for pest activity detection. The above sensors provide the foundation for the servos to activate for ventilation of fumes or the buzzers to activate pest deterrent mechanisms. This technology integrates with the Blynk IoT platform via the ESP32 microcontroller so that notifications could be communicated in real-time to mobile devices to inform the user if environmental conditions were breached in grain storage or to provide data on the internal conditions or any pest activity. The remoteness of the systems also makes observing environmental condition warnings even easier compared to having to visit each site. Once fully operational, it is perceived that users will be able to note changes in the condition of the crop they are monitoring without needing physical location access, therefor streamlining the user. The testing shows that once the environmental threshold was breached, the system was capable of responding consistently within 1-2 seconds. Also, the order of message relay, sensor data being communicated and actuators being activated were coherent across the tests, demonstrating the reliability of the outcomes. The inexpensive, readily available components used in the study allowed for an economically viable system, which is particularly attractive for rural use and smallholder farmers who desire to use this system.

In summary, this systems provides a clear delineation between traditional methods to store grain and the potential of smart systems, all whilst saving storage space.

## VI. Conclusion and Future Work

This paper describes a low-cost, Smart Crop Storage Monitoring System based on the Arduino platform. This project targets post-harvest losses for smallholder farmers. The system allows for monitoring environmental factors that are critical in crop quality post-harvest. The environmental conditions are measured (using DHT22, MQ135 and ultrasonic) to obtain key environmental factors, including, temperature, humidity, air quality and pest infestation. The system is aided with buzzers and servo motors for immediate responsible actions. The use of the Blynk IoT platform allow for real-time environmental data visualization and alerts to smartphones. The project is designed to be cost effective (less than USD 50 in total) and also available, accessible and reliable for rural intended audiences and resource-limited situations. Future work can include the addition of LoRa communication module to provide long-range connectivity to connect to solar-powered systems with no internet connectivity. Future works can also include integration of AI-based predictive analytics which can aid in the early warnings of changes in environmental hazards. The inclusion of solar to power this system will assist in being efficient and sustainable. The inclusion of a SMS module would also aid in accessibility where smartphone connectivity may be limited. Future improvement works will augment this system and desires for it to be high-level of intelligent, scalable and viable for wide-use in agricultural set-ups.

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